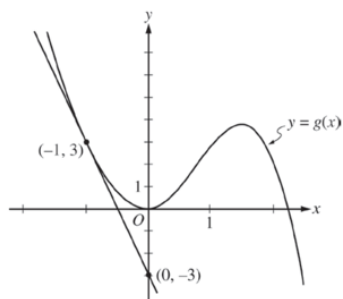




2.9 The Quotient Rule

2.10 Finding the Derivatives of Tangent, Cotangent, Secant, and Cosecant Functions

Ex 1:



The figure above shows the graph of the function g and the line tangent to the graph of g at $x = -1$. Let h be the function given by $h(x) = e^x \cdot g(x)$. What is the value of $h'(-1)$?

- (A) $\frac{9}{e}$ (B) $-\frac{3}{e}$ (C) $-\frac{6}{e}$ (D) $-\frac{6}{e} - \frac{3}{e^2}$ (E) -6

$$\begin{aligned} h'(x) &= e^x \cdot g'(x) + e^x \cdot g(x) \\ h'(-1) &= e^{-1}(g'(-1) + g(-1)) \\ h'(-1) &= e^{-1}(-6 + 3) = -\frac{3}{e} \end{aligned}$$

Quotient Rule

$$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{(g(x))^2}$$



Ex 2: If $h(x) = \frac{x}{3 \cos x + 1}$, find $h'(\pi)$.

$$\begin{aligned} h'(x) &= \frac{(3 \cos x + 1)(1) - (x)(-3 \sin x)}{(3 \cos x + 1)^2} \\ h'(x) &= \frac{3 \cos x + 1 + 3x \sin x}{(3 \cos x + 1)^2} \\ h'(\pi) &= \frac{3 \cos \pi + 1 + 3\pi \sin \pi}{(3 \cos \pi + 1)^2} = -\frac{3}{4} \end{aligned}$$

Ex 3: Write a tangent line to $h(x) = \frac{x}{3 \cos x + 1}$ at $x = \pi$.

$$\begin{aligned} h(\pi) &= \frac{\pi}{3 \cos \pi + 1} = -\frac{\pi}{2} \\ y + \frac{\pi}{2} &= -\frac{3}{4}(x - \pi) \end{aligned}$$



Not all quotients must be differentiated with the quotient rule. Sometimes it is easier or quicker to REWRITE first!

Ex 4: Which equations **must** be differentiated with the quotient rule?

- a) $f(x) = \frac{5}{x}$
b) $y = \frac{x}{5}$
(C) $y = \frac{e^x}{x+1}$
d) $g(x) = \frac{x+1}{x}$
e) $y = \frac{\sin x}{\cos x}$

Watch out for STDs!! (single term denominators)

Any quotient with a STD can be rewritten and differentiated WITHOUT the quotient rule.



Ex 5: Find the derivative.

a) $y = \tan x$

$$y = \frac{\sin x}{\cos x}$$

$$y' = \frac{(\cos x)(\cos x) - (\sin x)(-\sin x)}{(\cos x)^2}$$

$$y' = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x$$

b) $y = \cot x$

$$y = \frac{\cos x}{\sin x}$$

$$y' = \frac{(\sin x)(-\sin x) - (\cos x)(\cos x)}{(\sin x)^2}$$

$$y' = \frac{-\sin^2 x - \cos^2 x}{\sin^2 x} = \frac{-1}{\sin^2 x} = -\csc^2 x$$

c) $y = \sec x$

$$y = \frac{1}{\cos x}$$

$$y' = \frac{(\cos x)(0) - (1)(-\sin x)}{(\cos x)^2}$$

$$y' = \frac{\sin x}{\cos^2 x} = \frac{\sin x}{\cos x} \cdot \frac{1}{\cos x} = \tan x \sec x$$

d) $y = \csc x$

$$y = \frac{1}{\sin x}$$

$$y' = \frac{(\sin x)(0) - (1)(\cos x)}{(\sin x)^2}$$

$$y' = \frac{-\cos x}{\sin^2 x} = -\frac{\cos x}{\sin x} \cdot \frac{1}{\sin x} = -\cot x \csc x$$

Trigonometric Derivatives

$$\begin{array}{lll} \frac{d}{dx}[\sin x] = \cos x & \frac{d}{dx}[\cos x] = -\sin x & \frac{d}{dx}[\tan x] = \sec^2 x \\ \frac{d}{dx}[\csc x] = -\cot x \csc x & \frac{d}{dx}[\sec x] = \tan x \sec x & \frac{d}{dx}[\cot x] = -\cot x \csc x \end{array}$$

A Helpful Chart

$\tan x$	$\sec x$	$\sec x$
$\cot x$	$\csc x$	$-\csc x$



*Cover the term you wish to differentiate. The derivative is the product of the remaining terms in the same row.

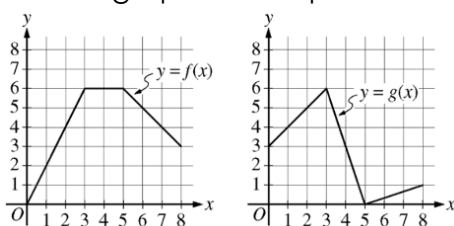
For example, the derivative of tangent is $\sec x \cdot \sec x = \sec^2 x$



$\sec x$

$\sec x$

Ex 6: The graphs of the piecewise linear functions f and g are shown.



If $m(x) = \frac{f(x)}{g(x)}$, find $m'(6)$ or state that it does not exist.

$$m'(x) = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{(g(x))^2}$$

$$m'(6) = \frac{g(6) \cdot f'(6) - f(6) \cdot g'(6)}{(g(6))^2} = \frac{\frac{1}{3} \cdot -1 - 5 \cdot \frac{1}{3}}{\left(\frac{1}{3}\right)^2} = -18$$

Ex 7: The functions $f(x)$ and $g(x)$ are differentiable. The table gives values of the functions and their first derivatives at selected values of x .

If $m(x) = \frac{g(x)}{f(x)}$, find $m'(7)$ or state that it does not exist.

x	0	2	4	7
$f(x)$	10	7	4	5
$f'(x)$	$\frac{3}{2}$	-8	3	6
$g(x)$	1	2	-3	0
$g'(x)$	5	4	2	8

$$m'(x) = \frac{f(x) \cdot g'(x) - g(x) \cdot f'(x)}{(f(x))^2}$$

$$m'(7) = \frac{f(7) \cdot g'(7) - g(7) \cdot f'(7)}{(f(7))^2}$$

$$m'(7) = \frac{6 \cdot 8 - 0 \cdot 6}{(5)^2} = \frac{48}{25}$$



Ex 8: Find the point(s) at which $f(x) = \frac{x^2}{x^2+1}$ has a horizontal tangent.

$$f'(x) = \frac{(x^2+1)(2x) - (x^2)(2x)}{(x^2+1)^2}$$

$$f'(x) = \frac{2x}{(x^2+1)^2} = 0$$

$$x = 0$$

$$f(0) = 0$$

Ex 9:

The function f is defined by $f(x) = \frac{x}{x+2}$. What points (x,y) on the graph of f have the property that the line tangent to f at (x,y) has slope $\frac{1}{2}$?

(A) $(0,0)$ only

(B) $(\frac{1}{2}, \frac{1}{5})$ only

☒ (C) $(0,0)$ and $(-4,2)$

(D) $(0,0)$ and $(4, \frac{2}{3})$

(E) There are no such points.

$$f'(x) = \frac{(x+2)(1) - (x)(1)}{(x+2)^2}$$

$$f'(x) = \frac{2}{(x+2)^2} = \frac{1}{2}$$

$$(x+2)^2 = 4$$

$$x = 0, -4$$

Ex 10:

The graph of $y = e^{\tan x} - 2$ crosses the x -axis at one point in the interval $[0, 1]$. What is the slope of the graph at this point?

(A) 0.606

(B) 2

(C) 2.242

☒ (D) 2.961

(E) 3.747

NORMAL FLOAT AUTO REAL Radian MP

Plot1 Plot2 Plot3

$Y_1 = e^{\tan(X)} - 2$

$Y_2 =$

$Y_3 =$

$Y_4 =$

$Y_5 =$

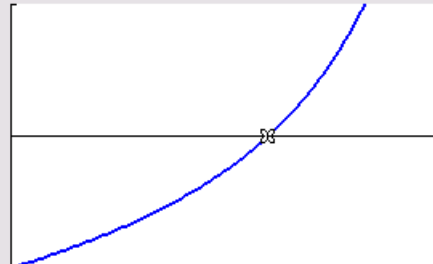
$Y_6 =$

$Y_7 =$

$Y_8 =$

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CALC ZERO

$Y_1 = e^{\tan(X)} - 2$



Zero

$X = 0.6061119$

$Y = 0$

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WINDOW

$X_{min} = 0$

$X_{max} = 1$

$X_{scl} = 1$

$Y_{min} = -1$

$Y_{max} = 1$

$Y_{scl} = 1$

$X_{res} = 1$

$\Delta X = 0.0037878787878788$

TraceStep = 0.007575757575...

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$\frac{d}{dx}(Y_1)|_{x=x}$

2.960912557